

3.2.2 Programming concepts

Content	Additional information
Use, understand and know how the following statement types can be combined in programs: • variable declaration • constant declaration • assignment • iteration • selection • subroutine (procedure/function).	The three combining principles (sequence, iteration/repetition and selection/choice) are basic to all high-level imperative programming languages. Students should be able to write programs using these statement types. They should be able to interpret and write algorithms that include these statement types. Students should know why named constants and variables are used.

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Use definite (count controlled) and indefinite (condition controlled) iteration, including indefinite iteration with the condition(s) at the start or the end of the iterative structure.	A theoretical understanding of condition(s) at either end of an iterative structure is required, regardless of whether they are supported by the language(s) being used. An example of definite (count controlled) iteration would be: <pre>FOR i ← 1 TO 5 ... Instructions here ... ENDFOR</pre> An example of indefinite (condition controlled) iteration with the condition at the start would be: <pre>WHILE NotSolved ... Instructions here ... ENDWHILE</pre> Examples of indefinite (condition controlled) iteration with the condition at the end would be: <pre>REPEAT ... Instructions here ... UNTIL Solved</pre> <pre>DO ... Instructions here ... WHILE NotSolved</pre>
Use nested selection and nested iteration structures.	An example of nested iteration would be: <pre>WHILE NotSolved ... Instructions here ... FOR i ← 1 TO 5 ... Instructions here ... ENDFOR ... Instructions here ... ENDWHILE</pre> An example of nested selection would be: <pre>IF GameWon THEN ... Instructions here ... IF Score > HighScore THEN ... Instructions here ... ENDIF ... Instructions here ... ENDIF</pre>
Use meaningful identifier names and know why it is important to use them.	Identifier names include names for variables, constants and subroutine names.